

tripleten

AI-Powered Customer Support Workflow

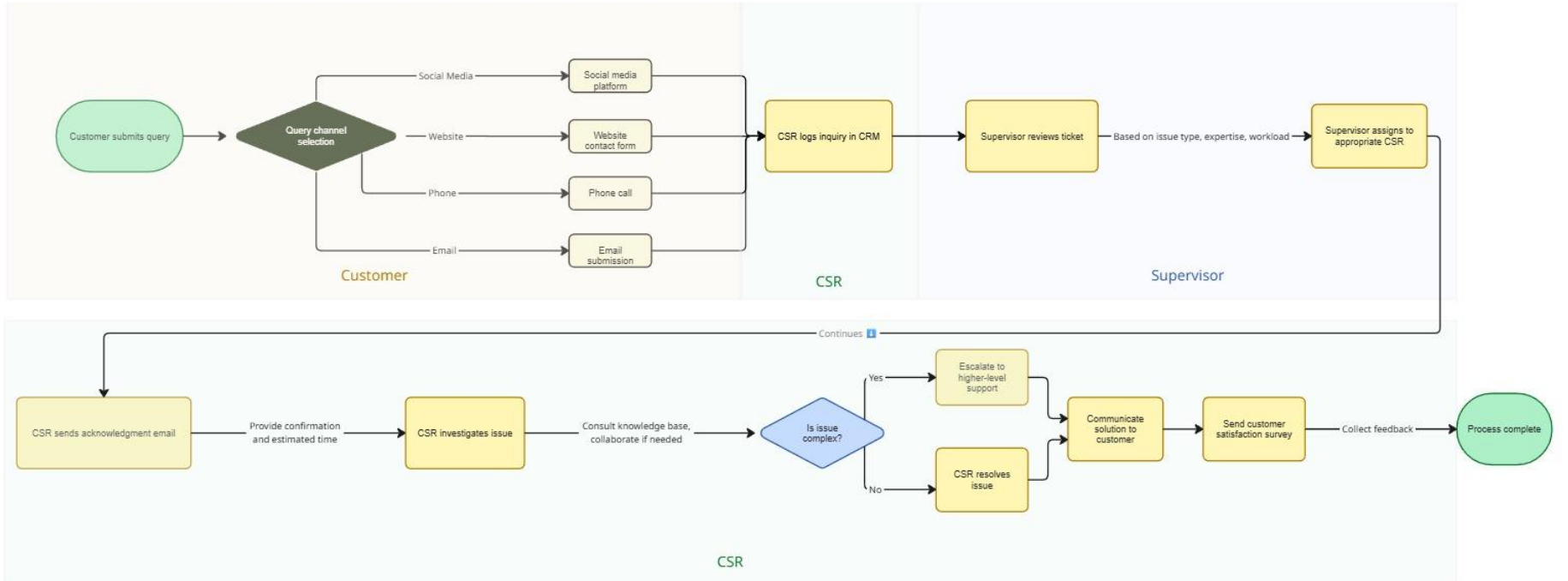
Brandon Robinson

Introduction

- **Overview of the Customer Support Process**
The current support process handles customer inquiries across email, phone, website forms, and social media. CSRs manually log each request, supervisors assign tickets based on the issue and workload, while CSRs investigate and resolve cases before sharing the final solution and gathering feedback.
- **Project Goal**
This project focuses on improving the speed and quality of the support workflow by using AI to automate routine steps, assist decision-making, and help deliver faster, more consistent resolutions.
- **AI-assisted analysis**
AI tools (ChatGPT) were used to organize and summarize complaint text, identify recurring customer frustrations, and support pattern recognition to group issues into clear problem themes.

Current Workflow Overview

Current-State Customer Support Workflow (Pre-AI)



Key Problems Identified*

1. **Long wait times and delays during initial contact:** Customers experience extended phone hold times, confusing automated menus, and slow responses before reaching a CSR, leading to frustration and repeated attempts to get help.
2. **Misrouted tickets and repeated transfers:** Inquiries are often assigned to the wrong CSR or department, causing customers to be transferred multiple times and repeat their issue, which slows resolution and increases workload.
3. **Lack of updates and slow investigation progress:** Customers report not receiving updates, unresolved issues after first contact, and incomplete final solutions. Manual investigation steps and unclear ownership extend resolution times.

**Complaint themes were summarized with the assistance of AI tools (ChatGPT) to identify recurring patterns.*

Proposed AI integration #1

AI Chatbot for Intake & Initial Triage

- Handles first-contact across website, email, and social channels.
- Collects basic issue details before a CSR is involved.
- Provides instant answers for common questions, reducing call volume.

Expected benefits and improvements

- Dramatically reduces wait times.
- Fewer customers stuck in phone trees or repeating information.
- CSRs receive cleaner, pre-structured inquiries.

Potential risks or limitations

- Chatbot may misinterpret complex issues.
- Customers with urgent or emotional concerns may prefer a human.
- Requires regular training to stay accurate.

Proposed AI integration #2

AI Ticket Classification & Smart Routing

- NLP model reads incoming inquiries and classifies:
 - Issue type
 - Priority level
 - Correct CSR/department
- Automatically creates or enriches tickets in CRM.

Expected benefits and improvements

- Reduces misrouted tickets and repeated transfers.
- Speeds up assignment and reduces supervisor workload.
- Improves first-time accuracy in routing, reducing rework.
- AI manages async channels so CSRs can handle phone calls end-to-end with fewer handoffs.

Potential risks or limitations

- Incorrect classification can still occur, especially early on.
- Needs historical labeled data to perform well.
- Supervisors must monitor and fine-tune model accuracy during rollout.

Proposed AI integration #3

AI-Assisted Investigation & Knowledge Support

- AI surfaces relevant knowledge articles and similar past cases.
- Provides quick summaries and suggested solution steps for CSRs.

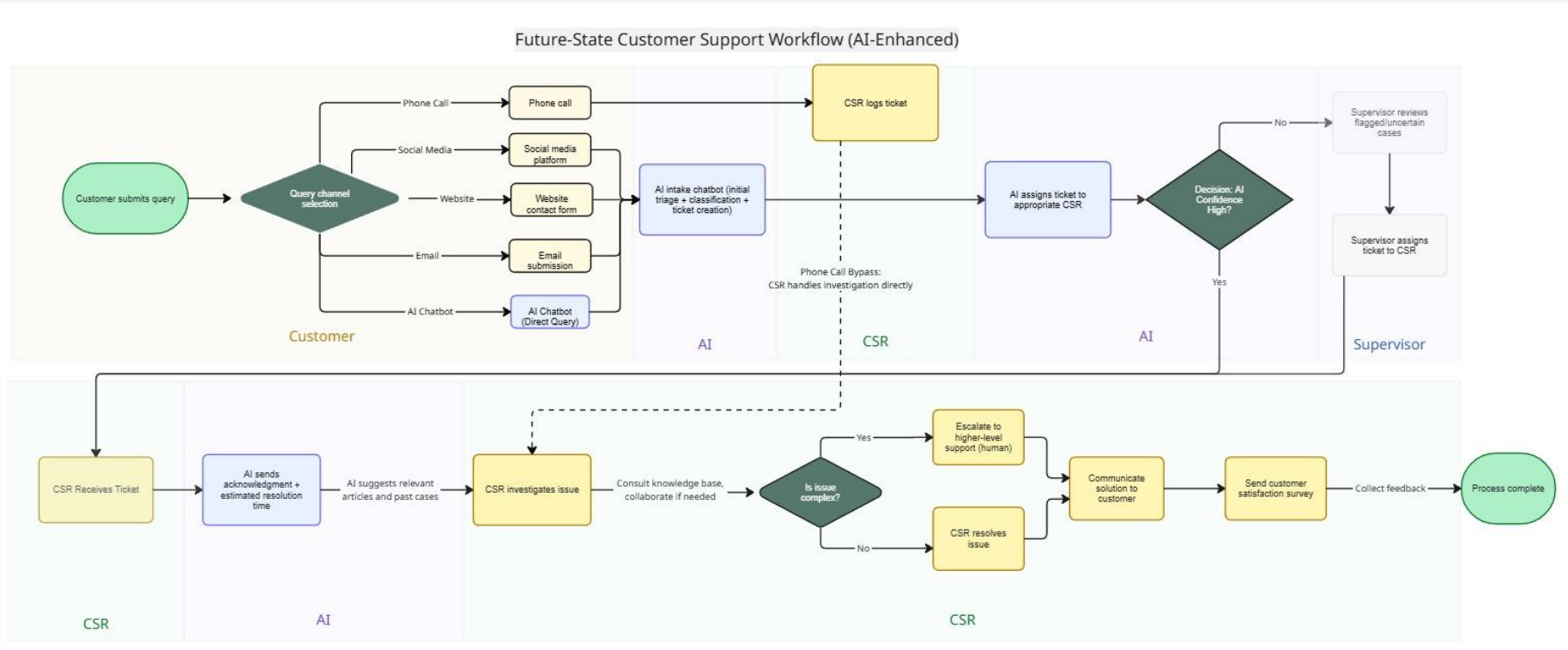
Expected benefits and improvements

- Faster CSR investigation and response times.
- More accurate, consistent solutions for customers.
- Helps newer CSRs ramp up faster.

Potential risks or limitations

- AI may reference outdated info if knowledge base isn't maintained.
- CSR must verify suggestions to avoid errors.

Updated Workflow with AI



Conclusion

The current support workflow creates friction for both customers and CSRs, resulting in long wait times, repeated transfers, and slow investigations.

Introducing AI Intake, smart ticket routing, and AI-assisted investigation reduces friction at each step while keeping humans in control of complex or sensitive cases.

Expected improvements include:

- Faster time to first response

- Fewer misrouted tickets and handoffs

- Clearer, better-structured cases for CSRs

With reduced logistics burden, CSRs can focus on problem-solving rather than ticket management.

With careful monitoring, phased rollout, and ongoing training, this AI-enhanced workflow can improve customer satisfaction and operational efficiency without sacrificing service quality.

Next Steps

1

Validate AI workflow design

Review the proposed workflow with stakeholders and confirm requirements.

2

Build and test prototypes

Create lightweight AI prototypes for intake, routing, and knowledge support. Test with real scenarios.

3

Plan phased rollout

Define training, monitoring, and metrics to deploy AI safely and gradually.

Optional AI Enhancements

AI Quality Monitoring

- AI analyzes call transcripts, chat logs, and CSR notes to flag recurring issues or bottlenecks.
- Generates weekly insight reports to guide performance improvements.
- Helps refine scripts, improve training, and reduce repeat contacts.

AI predicts issues before they escalate

- Detects patterns in customer behavior or product usage.
- Alerts CSRs or supervisors when certain cases are likely to require escalation.
- Helps prevent customer dissatisfaction before it surfaces.

Self-updating knowledge base

- AI identifies outdated articles and suggests revisions.
- Automatically drafts improved versions using recent cases and resolutions.
- Ensures CSRs and chatbots stay accurate and consistent.